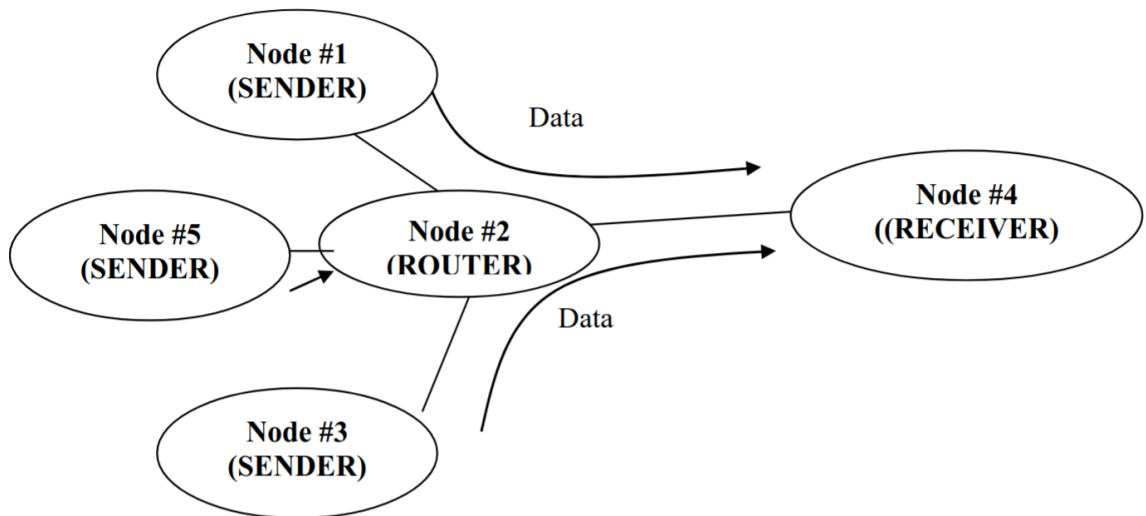


DCCN Lab Assignment - 5

Siddharth Narayan Mishra

123CS0189

1. Build a script to simulate the following network simulator scenario.
Apply TCP/FTP at node 1, 3 and UDP/CBR at node 5.



CODE

```
# Create Simulator
set ns [new Simulator]

# Define colors
$ns color 1 Blue
$ns color 2 Red
$ns color 3 Green

# Open NAM trace
set nf [open out.nam w]
$ns namtrace-all $nf

# Finish procedure
proc finish {} {
    global ns nf
    $ns flush-trace
```

```

    close $nf
    exec nam out.nam &
    exit 0
}

# Create nodes
set n0 [$ns node] ;# TCP sender 1
set n1 [$ns node] ;# Router
set n2 [$ns node] ;# TCP sender 2
set n3 [$ns node] ;# Receiver
set n4 [$ns node] ;# UDP sender

# Create links
$ns duplex-link $n0 $n1 1Mb 10ms DropTail
$ns duplex-link $n2 $n1 1Mb 10ms DropTail
$ns duplex-link $n4 $n1 1Mb 10ms DropTail
$ns duplex-link $n1 $n3 1Mb 10ms DropTail

# Set orientations (for clean NAM view)
$ns duplex-link-op $n0 $n1 orient right-down
$ns duplex-link-op $n2 $n1 orient right-up
$ns duplex-link-op $n4 $n1 orient left-down
$ns duplex-link-op $n1 $n3 orient right
$ns duplex-link-op $n1 $n3 queuePos 0.5

# ----- TCP Flow 1 -----
set tcp0 [new Agent/TCP]
$tcp0 set class_ 1
$ns attach-agent $n0 $tcp0

set sink0 [new Agent/TCPSink]
$ns attach-agent $n3 $sink0
$ns connect $tcp0 $sink0

set ftp0 [new Application/FTP]
$ftp0 attach-agent $tcp0

# ----- TCP Flow 2 -----

```

```
set tcp1 [new Agent/TCP]
$tcp1 set class_ 2
$ns attach-agent $n2 $tcp1
```

```
set sink1 [new Agent/TCPSink]
$ns attach-agent $n3 $sink1
$ns connect $tcp1 $sink1
```

```
set ftp1 [new Application/FTP]
$ftp1 attach-agent $tcp1
```

```
# ----- UDP Flow -----
set udp0 [new Agent/UDP]
$udp0 set class_ 3
$ns attach-agent $n4 $udp0
```

```
set null0 [new Agent/Null]
$ns attach-agent $n3 $null0
$ns connect $udp0 $null0
```

```
set cbr0 [new Application/Traffic/CBR]
$cbr0 set packetSize_ 500
$cbr0 set interval_ 0.005
$cbr0 attach-agent $udp0
```

```
# Schedule traffic
$ns at 0.5 "$ftp0 start"
$ns at 1.0 "$ftp1 start"
$ns at 1.5 "$cbr0 start"
```

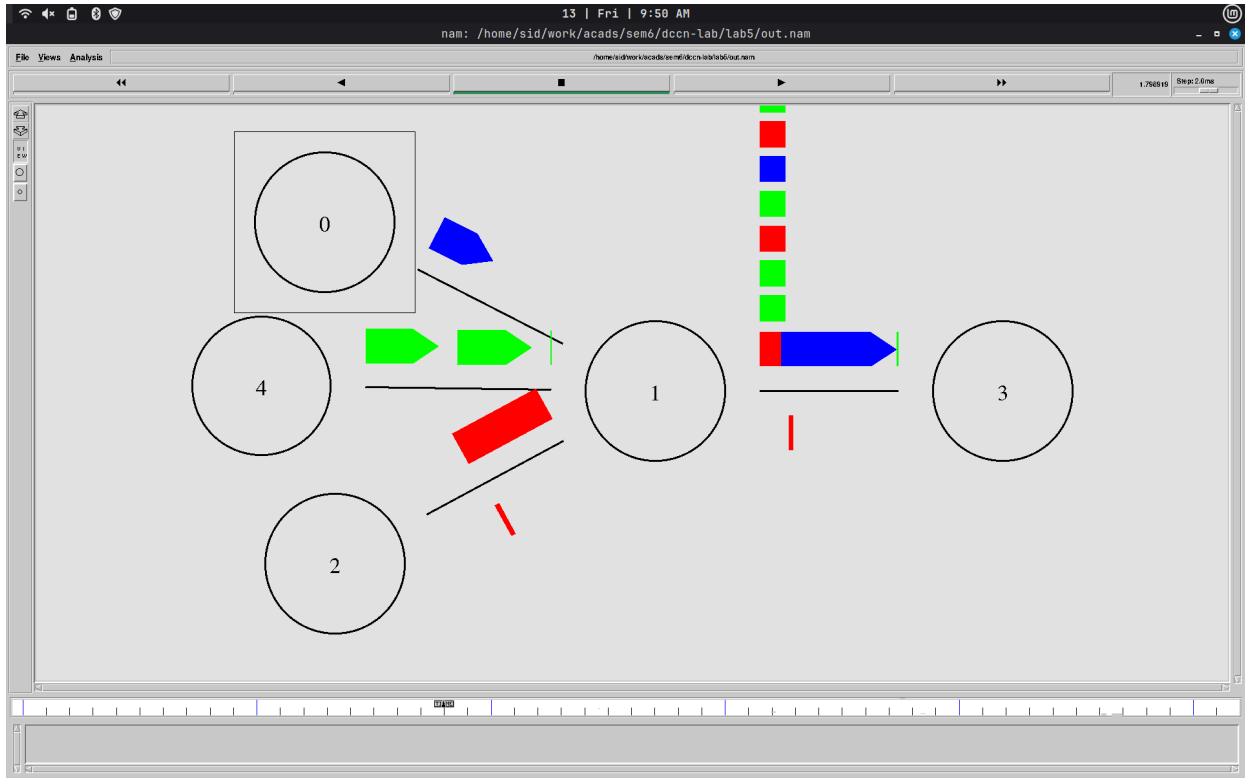
```
$ns at 4.0 "$cbr0 stop"
$ns at 4.5 "$ftp1 stop"
$ns at 5.0 "$ftp0 stop"
```

```
# End simulation
$ns at 6.0 "finish"
```

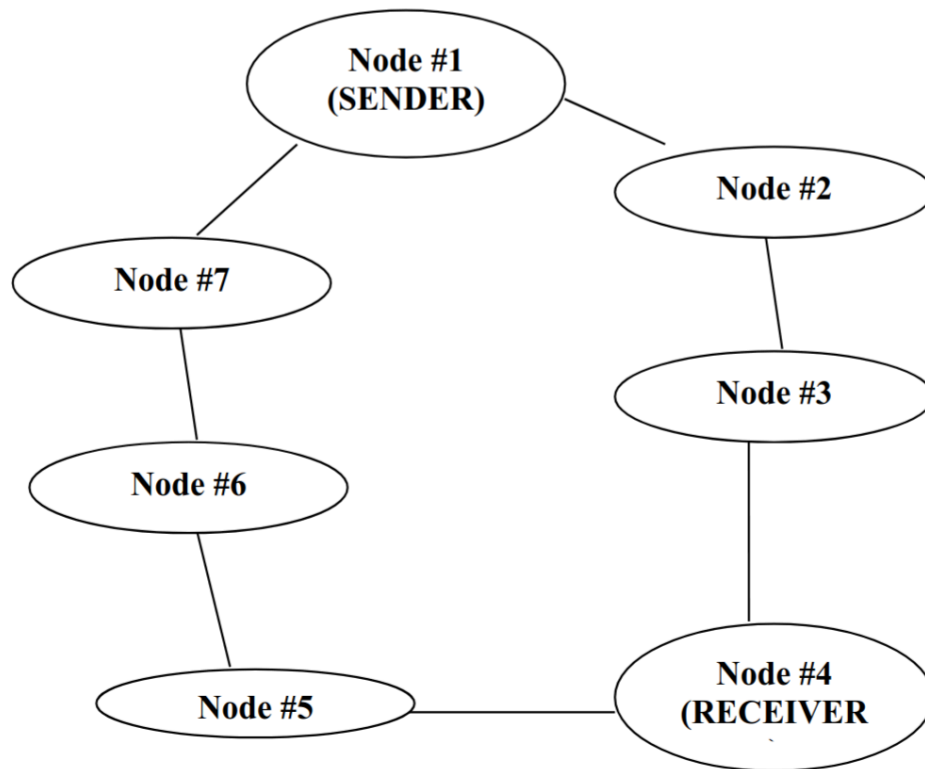
```
# Run
```

\$ns run

OUTPUT SCREENSHOT



2. Build a script to simulate following network simulator scenario



CODE

```
# Create Simulator
set ns [new Simulator]

# Define color
$ns color 1 Blue

# Open NAM trace
set nf [open out.nam w]
$ns namtrace-all $nf

# Finish procedure
proc finish {} {
    global ns nf
    $ns flush-trace
    close $nf
    exec nam out.nam &
    exit 0
}
```

```
}
```

```
# Create 7 nodes
```

```
set n0 [$ns node]
```

```
set n1 [$ns node]
```

```
set n2 [$ns node]
```

```
set n3 [$ns node]
```

```
set n4 [$ns node]
```

```
set n5 [$ns node]
```

```
set n6 [$ns node]
```

```
# Create ring links
```

```
$ns duplex-link $n0 $n1 1Mb 10ms DropTail
```

```
$ns duplex-link $n1 $n2 1Mb 10ms DropTail
```

```
$ns duplex-link $n2 $n3 1Mb 10ms DropTail
```

```
$ns duplex-link $n3 $n4 1Mb 10ms DropTail
```

```
$ns duplex-link $n4 $n5 1Mb 10ms DropTail
```

```
$ns duplex-link $n5 $n6 1Mb 10ms DropTail
```

```
$ns duplex-link $n6 $n0 1Mb 10ms DropTail ;# closes the cycle
```

```
# Orient links for clean NAM layout
```

```
$ns duplex-link-op $n0 $n1 orient right
```

```
$ns duplex-link-op $n1 $n2 orient right
```

```
$ns duplex-link-op $n2 $n3 orient down
```

```
$ns duplex-link-op $n3 $n4 orient left
```

```
$ns duplex-link-op $n4 $n5 orient left
```

```
$ns duplex-link-op $n5 $n6 orient up
```

```
$ns duplex-link-op $n6 $n0 orient right
```

```
# TCP Agent at n0
```

```
set tcp0 [new Agent/TCP]
```

```
$tcp0 set class_ 1
```

```
$ns attach-agent $n0 $tcp0
```

```
# TCP Sink at n3
```

```
set sink0 [new Agent/TCPSink]
```

```
$ns attach-agent $n3 $sink0
```

```
# Connect
$ns connect $tcp0 $sink0

# FTP Application
set ftp0 [new Application/FTP]
$ftp0 attach-agent $tcp0

# Start/Stop
$ns at 0.5 "$ftp0 start"
$ns at 4.5 "$ftp0 stop"

# Finish
$ns at 5.0 "finish"

# Run
$ns run
```

OUTPUT SCREENSHOT

